Event-Aware Representation Learning for Neutral Multi-Source News Summarization

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1 Project Idea

We propose a neutral multi-source news summarization system whose internal representation is explicitly organised around the *event* being reported rather than the *publisher* reporting it. Given several articles about the same story from outlets of differing political leaning, the model produces a single neutral summary; alongside the usual generation objective, its encoder is trained with a triplet loss that pulls together differently-framed reports of the same event and pushes apart same-publisher reports of different events.

The deliverable is a summarization model with an event-aware representation-learning objective, evaluated on how much event content it retains against how much publisher-specific information remains recoverable from its representations.

Before writing this proposal we profiled every corpus we intend to use, reimplemented the evaluation metric the relevant literature reports, and ran the core experiment once end to end. It works: the objective cuts publisher recoverability by 63% at unchanged ROUGE, but leaves the framing-bias metric essentially untouched. That gap is what the project is now about. The findings below are the substance of our motivation, not background detail.

2 Motivation

Nothing constrains what the encoder organises around. NeuS Lee et al. (2022) defines the neutral multi-news summarization task and shows that generation models can hallucinate politically biased content, not merely factually incorrect content. Its training signal, however, supervises only the output text. Independently, Baly et al. (2020) show that news encoders readily latch onto source-specific characteristics — a publisher's stylistic fingerprint — rather than the content a task actually needs. If a summarization encoder does the same, two failure modes follow: *framing leakage*, where a summary's tone shifts with which outlets happened to be sampled even though the text reads as neutral, and *neutrality by deletion*, where the model appears neutral only by dropping contested details instead of reconciling them.

Publisher identity is present and stance does not transfer. We verified the premise rather than assuming it. On the 9,282 articles of the NeuS corpus, a plain TF-IDF logistic probe recovers the publishing outlet at 17.8% accuracy across 229 outlets, against a 9.2% majority baseline. That is roughly twice the corresponding baseline, so outlet identity is plainly written into the words. Stance behaves in the opposite way. Predicting an article's left/centre/right leaning reaches only 40.2% when outlets are shared between train and test, against a 37.3% majority baseline; and when no outlet appears on both sides it falls to 35.4%, *below* the 46.0% obtainable by always predicting the most frequent class and barely above the 33.3% chance level. Lexical features therefore encode *which outlet wrote this* and not a transferable notion of ideology — the signal a model trained on this data will reach for first is exactly the one we want its representation to discard. This reproduces Baly et al.'s argument on the very corpus we intend to generate from, and it is precisely the confound an event-aware objective should remove.

The field's neutrality metric rewards saying less. The framing-bias metric introduced with NeuS scores a generation by summing NRC-VAD Mohammad (2018) arousal over the tokens it uses that the neutral reference does not. We reimplemented it — the released notebook does not run — and measured it on gold data. Across all 9,282 gold articles the metric correlates 0.88 with the number of calibrated tokens, and once divided by that count the affect rate is flat: 0.1767, 0.1763 and 0.1792 for left, centre and right respectively, against a standard deviation of 0.081. The same ≈ 0.177 appears on WCEP-10 Ghalandari et al. (2020), a different corpus from a different period. The metric separates partisan from centrist text only because partisan text diverges more in vocabulary from the reference, not because that vocabulary carries more affect.

The consequence is direct: a model can lower its framing-bias score simply by generating fewer novel tokens. The headline metric is blind to neutrality by deletion — one of the two failure modes the task exists to prevent. To our knowledge this has not been reported.

The standard split cannot test generalisation. NeuS's released train/validation/test split is random over stories. Recovering the publisher labels, we find that 92 of the 99 test outlets also occur in training. Any claim about publisher leakage measured on this split is a claim about outlets the model has already seen.

3 Preliminary Results

The data pipeline, the metric, and a first end-to-end experiment are already built and run. Table 1 summarises what each corpus can support; Table 2 is the result that motivates the rest of the project.

Corpus	Event	Publisher	Neutral
NeuS (3,095 stories)	yes	yes	yes
AllSides (37,554 arts.)	no	yes	no
WCEP-10 (10,200 clus.)	yes	no	partial
WNC (181k sent. pairs)	no	no	partial

Table 1: An event-aware triplet objective needs all three labels on one example. Only NeuS supplies them together, and it is the smallest of the four.

Representation	Pub.	Stance	R-1
TF-IDF (lexical floor)	17.8	40.2	—
BART, generation loss only	21.5	48.3	38.63
BART + event-aware triplet	**7.9**	38.6	38.72
majority baseline	9.2	37.3	—

Table 2: Logistic probes on frozen encoder representations of 9,282 articles; publisher is a 229-way problem. Training for summarization makes the publisher *more* recoverable than raw word counts. The triplet objective drives it below the majority baseline while ROUGE-1 is unchanged.

Setup. BART-base fine-tuned on the 2,452 NeuS training examples for five epochs at fp16 on a single RTX 2080 Ti, once with the generation loss alone and once with $\lambda = 1.0$, $m = 1.0$. Six minutes per run. Probes are logistic regressions on the pooled encoder state, evaluated on a held-out article split.

Publisher identity is learned, then removed. A TF-IDF probe recovers the outlet at 17.8% over 229 classes against a 9.2% majority baseline, so outlet identity is written into the words. Fine-tuning for summarization makes this *worse*, not better: the baseline encoder reaches 21.5%. The summarization objective actively organises the representation around the publisher. Adding the triplet term drops it to 7.9% — below the majority baseline, i.e. effectively unrecoverable — a 63% relative reduction, with ROUGE-1 unchanged at 38.7 versus 38.6 and identical mean output length.

This is not representation collapse: a collapsed encoder would put the triplet loss at the margin, 1.0, and it converges to 0.047.

But neutrality does not follow. The framing-bias metric barely moves — $1.732 \rightarrow 1.701$ raw, $0.1774 \rightarrow 0.1751$ per calibrated token, on identical generation lengths, so this is a genuine non-effect rather than the length artifact described below. **Removing publisher information from the encoder does not by itself make the output more neutral.** Representation-level leakage and surface-level framing bias are separable, and the project's central question becomes what closes that gap.

A caveat we will report. Stance recoverability falls alongside publisher ($48.3 \rightarrow 38.6$). Because stance is an outlet-level property in this corpus, the objective cannot strip publisher identity while preserving ideology — the inverse of what Baly et al. (2020) achieve, and a real limitation of applying their technique in this direction.

4 Background and Related Work

NeuS Lee et al. (2022) introduces the task, the AllSides-derived corpus of left/centre/right article triples with human-written neutral "roundup" summaries, and the framing-bias metric. It fine-tunes BART Lewis et al. (2020) on the concatenated sources and reports that a multi-task auxiliary objective over the encoder helps — but the auxiliary task is classification, and nothing ties the representation to event identity.

We Can Detect Your Bias Baly et al. (2020) predicts article-level political ideology and uses a triplet objective to strip publisher identity from a representation while preserving ideology. It also contributes the media-based split protocol, in which whole outlets are held out. We invert the objective: preserve *event* identity while removing *source* information.

Multi-document event relation graphs Lei and Huang (2024) target the same framing-bias mitigation goal on the same corpus by building an explicit event relation graph and reasoning over it with an LLM. Our approach is the representation-level counterpart: rather than externalising event structure into a graph, we constrain the encoder's latent space to be organised by event. We regard this as the closest prior work and will compare against it directly.

I-JEPA Assran et al. (2023) predicts one view's latent representation from another rather than reconstructing raw content; this motivates our label-efficient pretraining extension, where the "views" are two outlets' reports of the same event.

DiverseSumm Huang et al. (2024) proposes QA-based evaluation of whether a multi-document summary retains information from its sources.

5 Contributions

1. **An event-aware triplet objective** for multi-source neutral summarization, in which the positive shares an event with the anchor and the negative shares a *publisher*. The shared publisher is the mechanism: it is the only configuration in which satisfying the margin requires the encoder to stop relying on outlet style. This addresses the first shortcoming — that standard training leaves the encoder's organising principle unconstrained — and a first run already shows a 63% relative cut in publisher recoverability at unchanged ROUGE.
2. **A length-normalised framing-bias metric**, together with the evidence that the standard metric is a length proxy. We will report both, and show what conclusions change. This addresses the third shortcoming, and makes neutrality-by-deletion visible rather than rewarded.
3. **Evidence that representation leakage and output framing bias are separable.** Our first experiment removes publisher information from the encoder without

moving the framing-bias metric at all, on identical output lengths. Whether that gap can be closed — and what it says about representation-level interventions for surface-level bias — is the question the rest of the project answers.

4. **A publisher-disjoint evaluation protocol** for this corpus: a stance-stratified split in which no outlet appears on two sides, plus a leakage probe measured on both the standard and the disjoint split. This addresses the fourth shortcoming — that the released split cannot ask the generalisation question.
5. **A reproducible data pipeline.** The released NeuS split files have their publisher and stance labels stripped. We recover them at 100% coverage by normalising headlines against the union of all three crawl files — the obvious single-file join silently misses 13% of examples. We will release this alongside our code.

6 Proposed Methodology

Backbone. A pretrained sequence-to-sequence model (BART-large). The k articles for an event are concatenated into a single source sequence and decoded into a neutral summary under the usual token-level cross-entropy loss against the human-written roundup.

Event-aware triplet objective. For an anchor article a, we sample a positive p — a different outlet's report of the same event — and a negative n — the same outlet's report of a different event. Each is passed through the shared encoder and pooled at the EOS position to give $f(\cdot)$. The training signal is

$$\mathcal{L} \;=\; \mathcal{L}_{\text{gen}} + \lambda \cdot \max\big(0,\; \|f(a)-f(p)\| - \|f(a)-f(n)\| + m\big),$$

with margin m and weight λ swept on the validation set. Because a and n always share a publisher, the only way to satisfy the margin consistently is to stop encoding publisher style and organise around the event instead.

Structurally this attaches at the same point as NeuS's own auxiliary-task hook, which adds a weighted auxiliary loss over a pooled encoder state; we replace their binary classification head with the triplet term and reimplement on a current transformers stack rather than their pinned 2021 fork.

Sampling and training details. Triples are mined offline: for each of the 9,282 articles we draw two positives and two negatives, discarding anchors whose outlet appears in only one story, which yields 18,368 triples. The triplet term is fed as a second stream alongside the generation batch, mirroring the reference implementation's separate auxiliary-input path, so the two losses need not share a batch size. We start from the hyperparameters NeuS reports — learning rate 3×10^{-5}, effective batch 16, source length 512, target length 250 — and sweep only λ and m. Training is short: 2,452 examples over five epochs is roughly 3,000 optimiser steps, under an hour of single-GPU time per configuration, which is what makes the ablation grid and the multi-seed reporting affordable.

Extension: cross-source pretraining. Following the JEPA principle, we pretrain by predicting one outlet's representation of an event from another outlet's representation of the same event, using unlabelled same-event pairs from WCEP-10 — which needs neither publisher labels nor neutral targets, and so suits this objective exactly. The question is whether the decoder can then be fine-tuned on a fraction of the labelled summaries and still match a fully supervised model. This extension is contingent on compute and will be dropped before the core result if necessary.

7 Dataset

Our primary corpus is the NeuS release Lee et al. (2022): 3,095 AllSides stories, each pairing three articles (one left, one centre, one right) with a human-written neutral roundup, spanning 2016–2021 and 250 distinct outlets. It is the only corpus available to us that carries an event grouping, a publisher label and a neutral target on the same example. From it we derive 9,282 publisher-labelled articles and 18,368 training triples; 153 of the 250 outlets appear in at least two distinct stories, which makes 9,188 of 9,285 article slots usable as a same-publisher negative.

Two properties we flag honestly. First, AllSides publishes an *excerpt* per outlet rather than the full article — a hard cap of 95 words, median 85 — so a three-source input is ≈241 words and fits comfortably within BART's 1,024-token window, but there is correspondingly little event content to retain. Second, at 2,452 training examples the corpus is small for fine-tuning BART-large; we will report mean and spread over at least three seeds rather than single runs.

We use two further corpora in supporting roles. **WCEP-10** Ghalandari et al. (2020) provides 10,200 genuine multi-document event clusters of ≈9 articles each — far better event structure and scale than NeuS — but its preprocessing stripped the publisher labels and its 25-word targets were never written to be neutral, so we use it only for the pretraining extension. **AllSides / Article-Bias-Prediction** Baly et al. (2020) supplies 37,554 publisher-labelled articles for additional negatives and for its prebuilt media-based split. We note that its bias labels are outlet-level: only 7 of its 491 sources ever carry more than one label, so ideology prediction and publisher prediction are near-identical tasks there and we will not report them as independent axes.

8 Evaluation Plan

Summary quality. ROUGE-1/2/L against the human roundups, alongside two training-free baselines the model must beat: concatenating all three source articles, and copying the centrist source. The latter matters — on the raw framing-bias metric, copying the centre article already scores below both partisan articles at zero training cost.

Framing bias, both ways. The NeuS metric as published, and our length-normalised variant (arousal per calibrated affect-bearing token), reported side by side with mean generation length. A run that improves the raw score without improving the normalised rate has learned to say

less, and we will label it as such. We additionally track MPQA strong-subjectivity density Wilson et al. (2005), the one lexical signal in our audit that is *not* flat under length ($0.0633/0.0557/0.0631$ per calibrated token for left/centre/right).

Publisher leakage. A logistic probe on frozen encoder representations, predicting the source outlet; lower is better. Measured on the random split *and* on our stance-stratified publisher-disjoint split, with a TF-IDF probe as the lexical floor so that we can tell a representation-level improvement from a vocabulary-level one. The comparison of interest is generation-only training versus triplet training at matched ROUGE.

Event retention. A DiverseSumm-style QA check Huang et al. (2024), scoped down honestly: with 85-word source excerpts there is limited room for questions answerable from one source and not another, so we treat this as a secondary content-preservation check rather than a headline result.

Ablations. $\lambda = 0$ (generation only); triplet with negatives sampled from a *different* publisher, which removes the adversarial pressure and should recover most of the leakage; and a publisher-shuffled control.

9 Feasibility and Risks

The compute is modest: the full experimental grid — baseline, triplet sweep, ablations, and reruns on the disjoint split — is roughly 20–25 single-GPU hours at 16 GB, which our cluster allocation covers comfortably. The data pipeline, the metric, and the probes are CPU-only and already run.

Three risks we are managing explicitly. **Small data:** 2,452 training examples will produce noticeable seed variance, so every reported number will be a mean over at least three seeds with spread. **Weak event content:** the 85-word source excerpts limit what an event-retention evaluation can claim, so we have demoted the QA check to a secondary measure rather than build the argument on it. **A null result is possible:** the triplet term may reduce publisher recoverability without improving summary neutrality, or may trade ROUGE for it. We regard that as a reportable outcome — combined with the metric critique it still constitutes a complete finding about what representation-level constraints can and cannot buy in this task.

10 Timeline

Weeks	Milestone
1–2	Data pipeline, splits, probe protocol *(complete)*
3–4	NeuS baseline reproduced; metric analysis written up
5–7	Triplet objective; λ/margin sweep; leakage probes on both splits
8–9	Ablations, seed variance, QA check
10–11	Cross-source pretraining extension (compute permitting)
12	Final report and code release

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